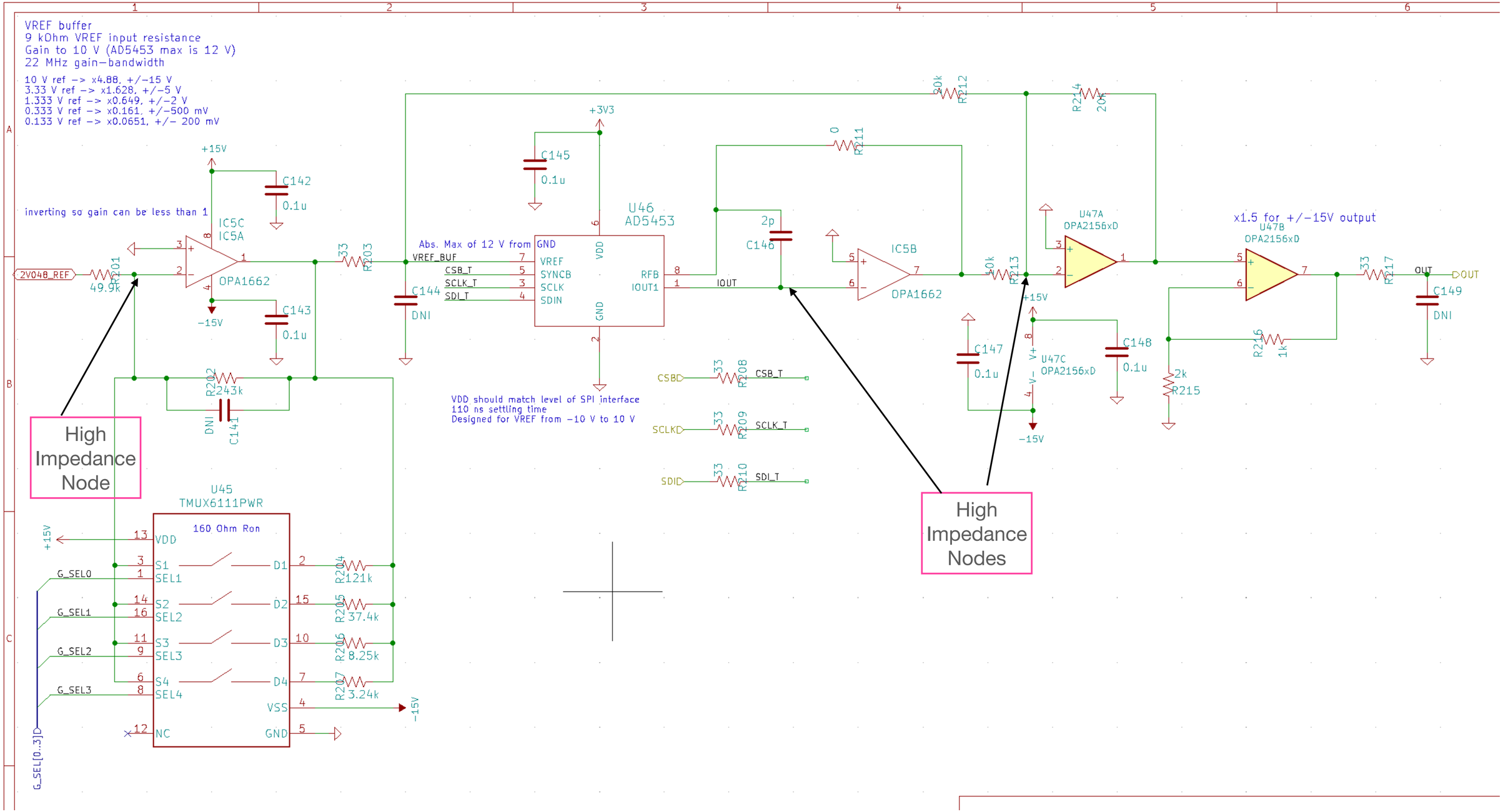
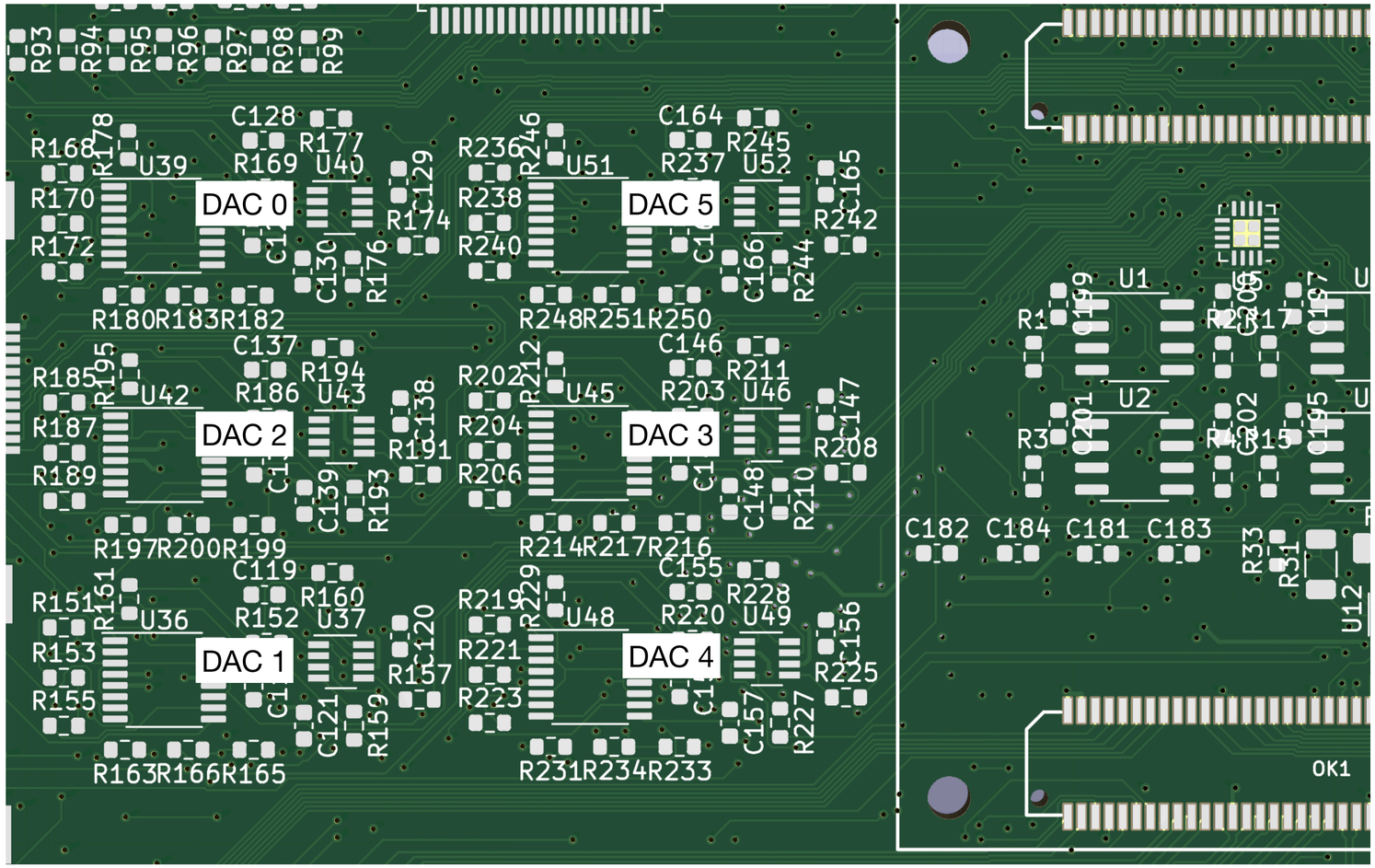
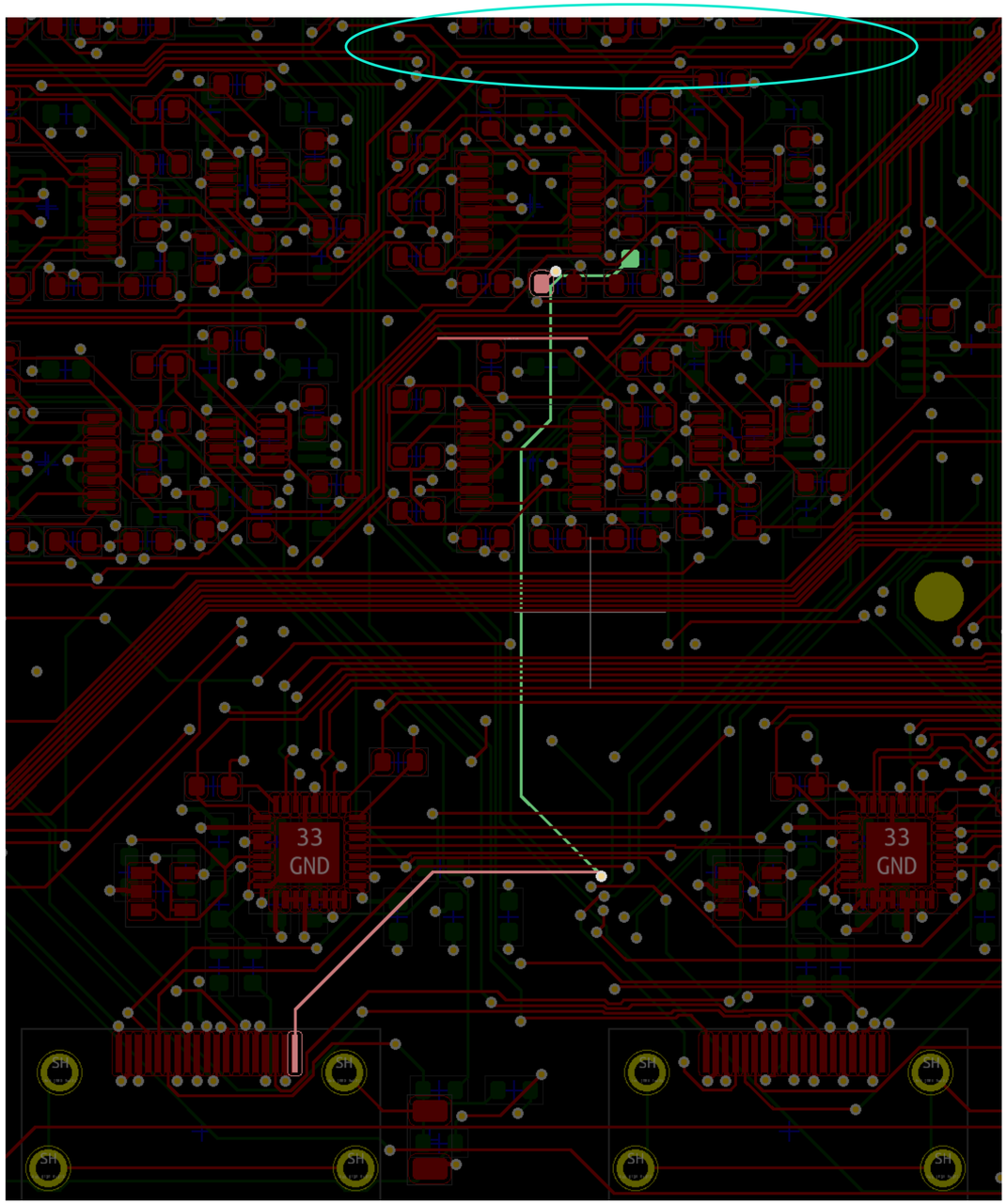

VREF buffer
9 kOhm VREF input resistance
Gain to 10 V (AD5453 max is 12 V)
22 MHz gain-bandwidth
10 V ref -> x4.88, +/-15 V
3.33 V ref -> x1.628, +/-5 V
1.333 V ref -> x0.649, +/-2 V
0.333 V ref -> x0.161, +/-500 mV
0.133 V ref -> x0.0651, +/- 200 mV
inverting so gain can be less than 1
2V048_REF
R201 49.9k
IC5C IC5A OPA1662
+15V
-15V
C142 0.1u
C143 0.1u
R202 243k
DNI C141
R203 33
C145 0.1u
+3V3
U46 AD5453
Abs. Max of 12 V from
VREF_BUF
CSB_T
SCLK_T
SDI_T
GND
VREF
SYNCB
SCLK
SDIN
VDD
RFB
IOUT1
C144 DNI
VDD should match level of SPI interface
110 ns settling time
Designed for VREF from -10 V to 10 V
CSB
SCLK
SDI
R208 33
R209 33
R210 33
CSB_T
SCLK_T
SDI_T
R211 0
2p C146
IOUT
IC5B OPA1662
R212 20k
R213 10k
R214 20k
U47A OPA2156xD
U47B OPA2156xD
x1.5 for +/-15V output
U47C OPA2156xD
C147 0.1u
C148 0.1u
R215 2k
R216 1k
R217 33
OUT
C149 DNI
High Impedance Node
High Impedance Nodes
U45 TMUX6111PWR
160 Ohm Ron
VDD
S1 SEL1
S2 SEL2
S3 SEL3
S4 SEL4
NC
D1 D2 D3 D4
VSS
GND
R204 121k
R205 37.4k
R206 8.25k
R207 3.24k
G_SEL0
G_SEL1
G_SEL2
G_SEL3
G_SEL[0..3]
+15V
-15V

Channel	Ripple	Notes
0	30	
1	30	
2	22	
3	130	Chan. 2 is most significant aggresor
4	34	
5	60	



High ripple are all the right-most DACs
Compare 3 and 4 as right-most DACs on the ends of the spectrum for copuling.

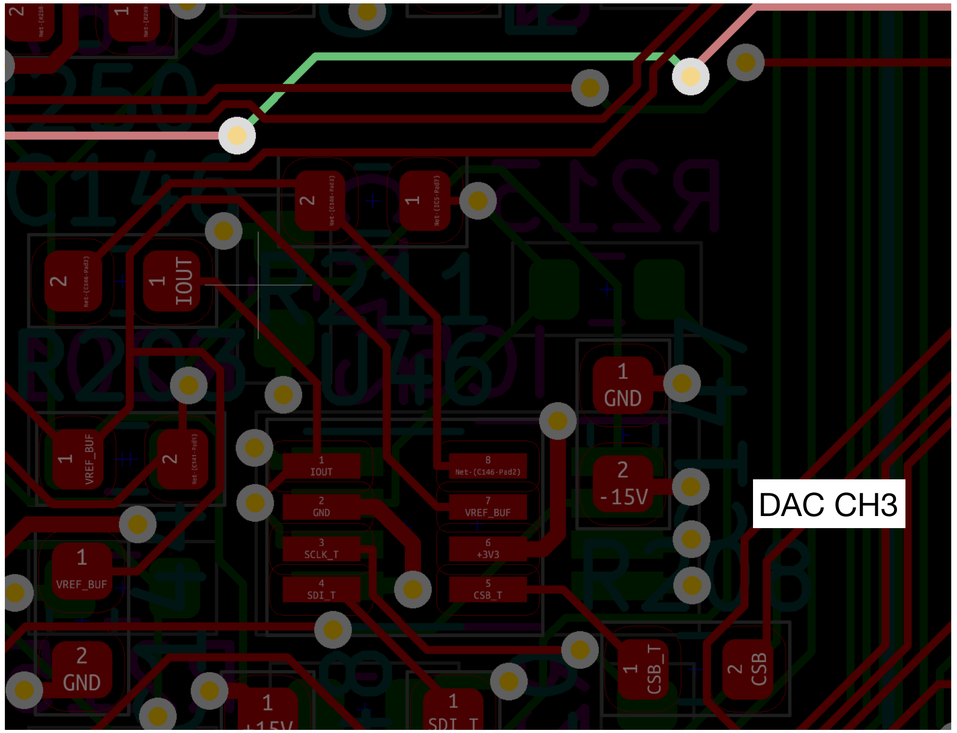
dac2 SPI lines (CSb and SDI)



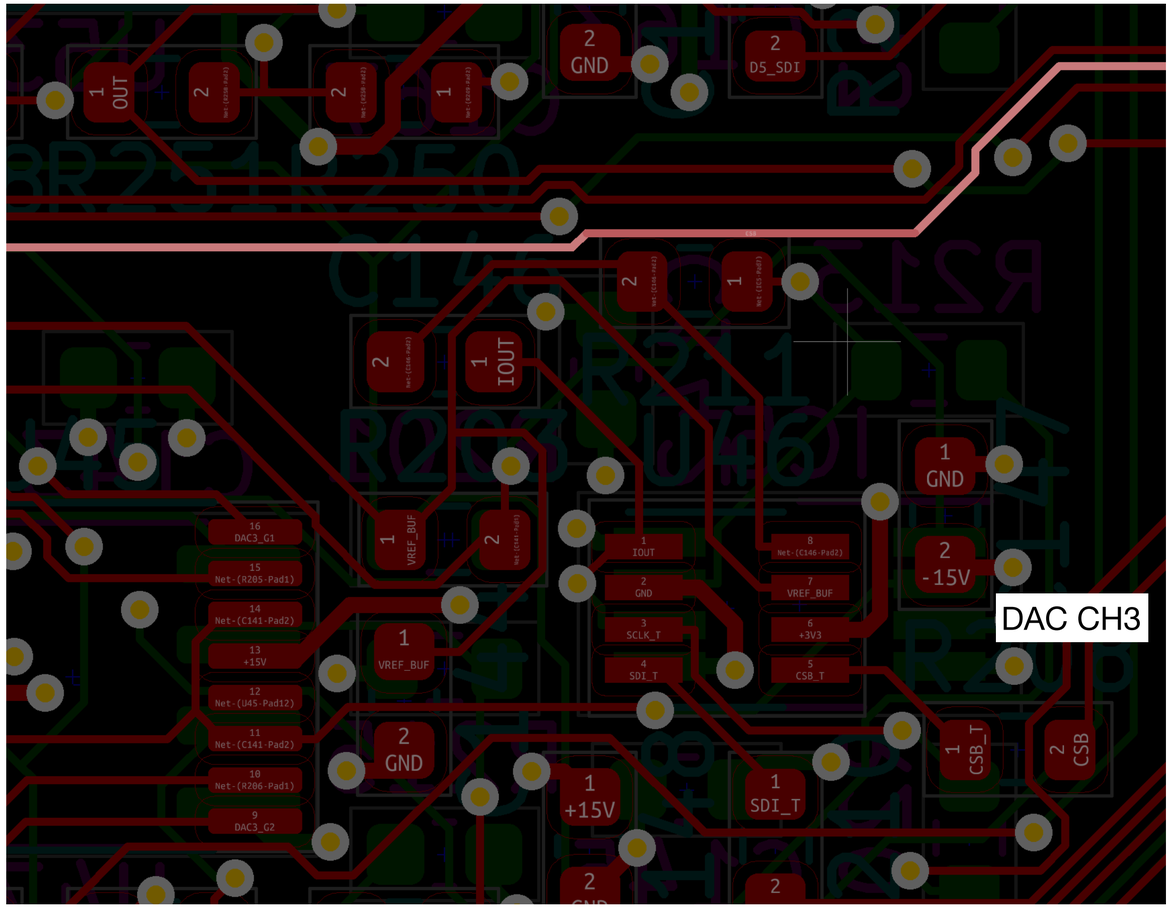
Highlighted Net: dac3/OUT

OUT does not intersect with dac2 SPI lines

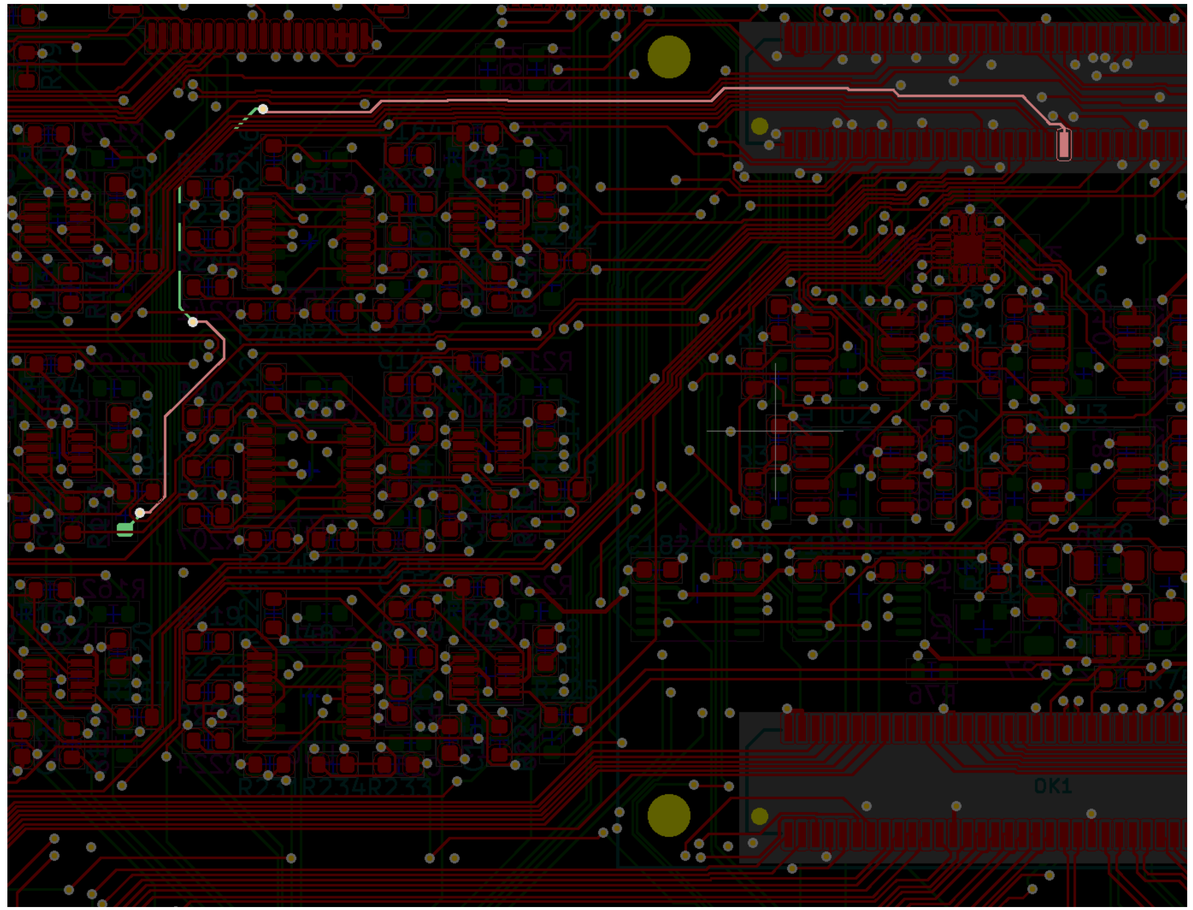
Highlighted Net: dac2/SDI
Very close to IOUT and the capacitively coupled IOUT that routes back to pin 8



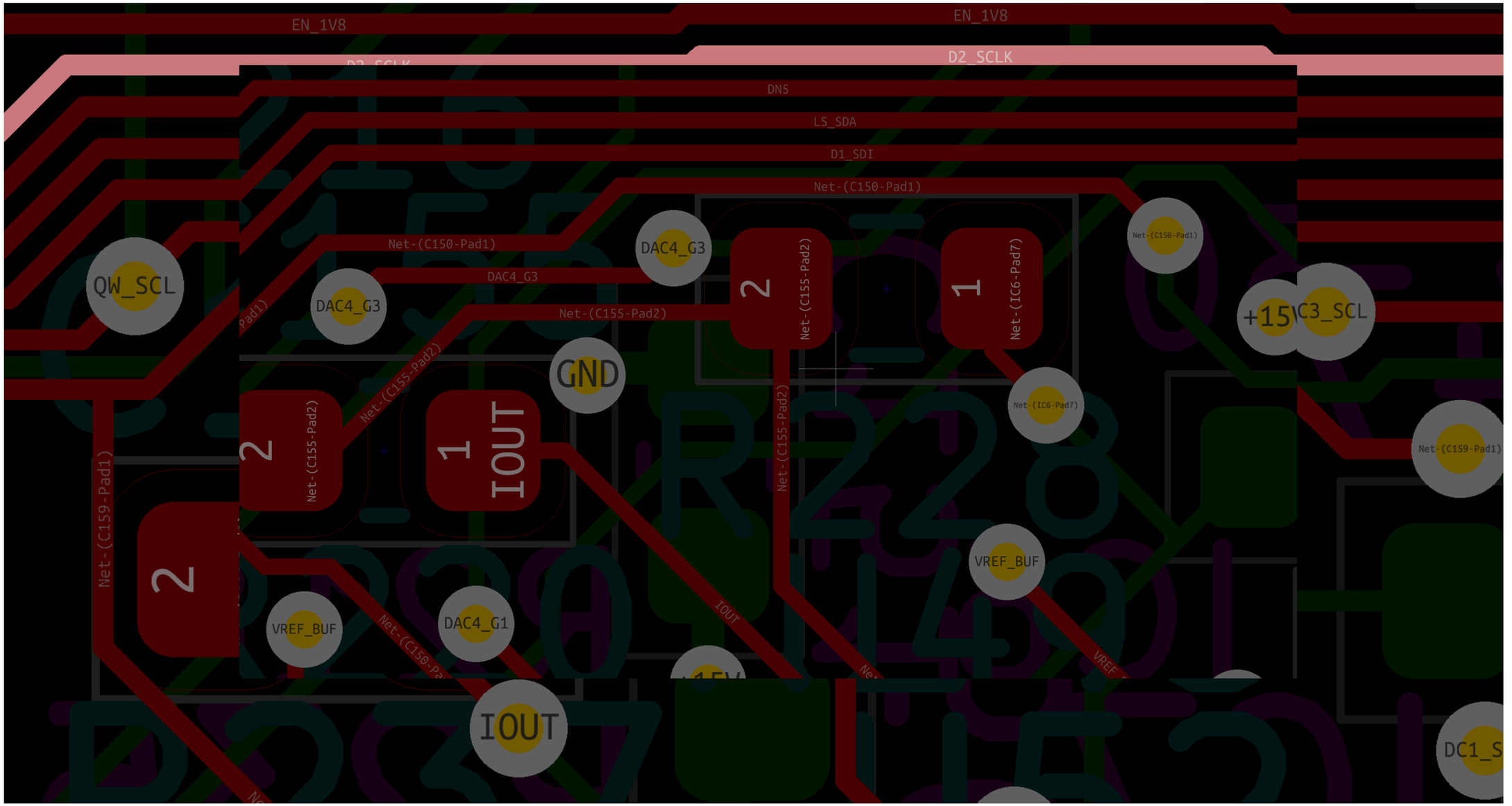
Highlighted Net: dac2/CSB
Very close to IOUT and the capacitively coupled IOUT that routes back to pin 8



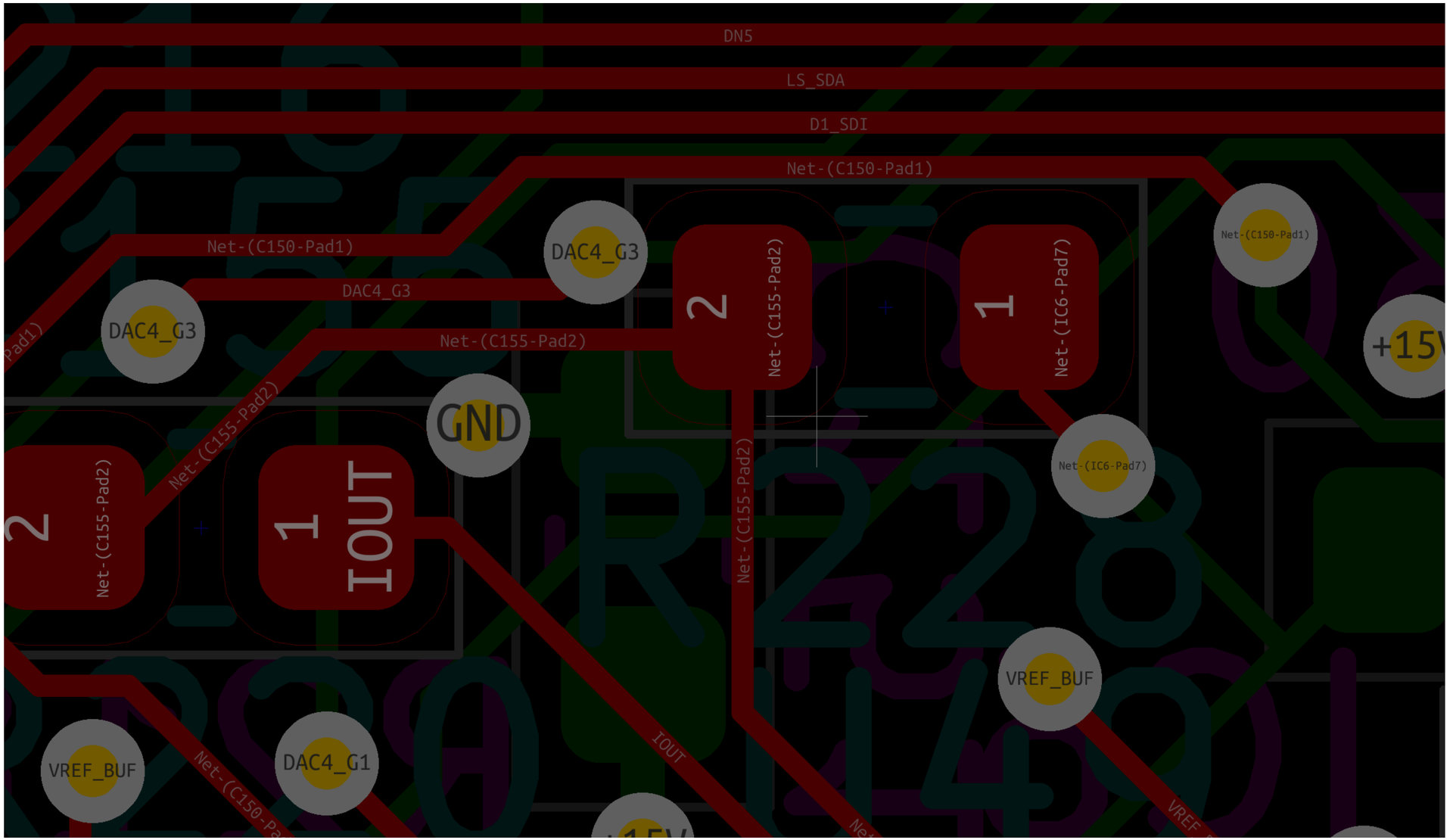
Highlighted Net: dac2/SCLK
This net does not interact with DAC3



Near IOUT of DAC5
QW_SCL is in between but then there are 4 consecutive SPI lines

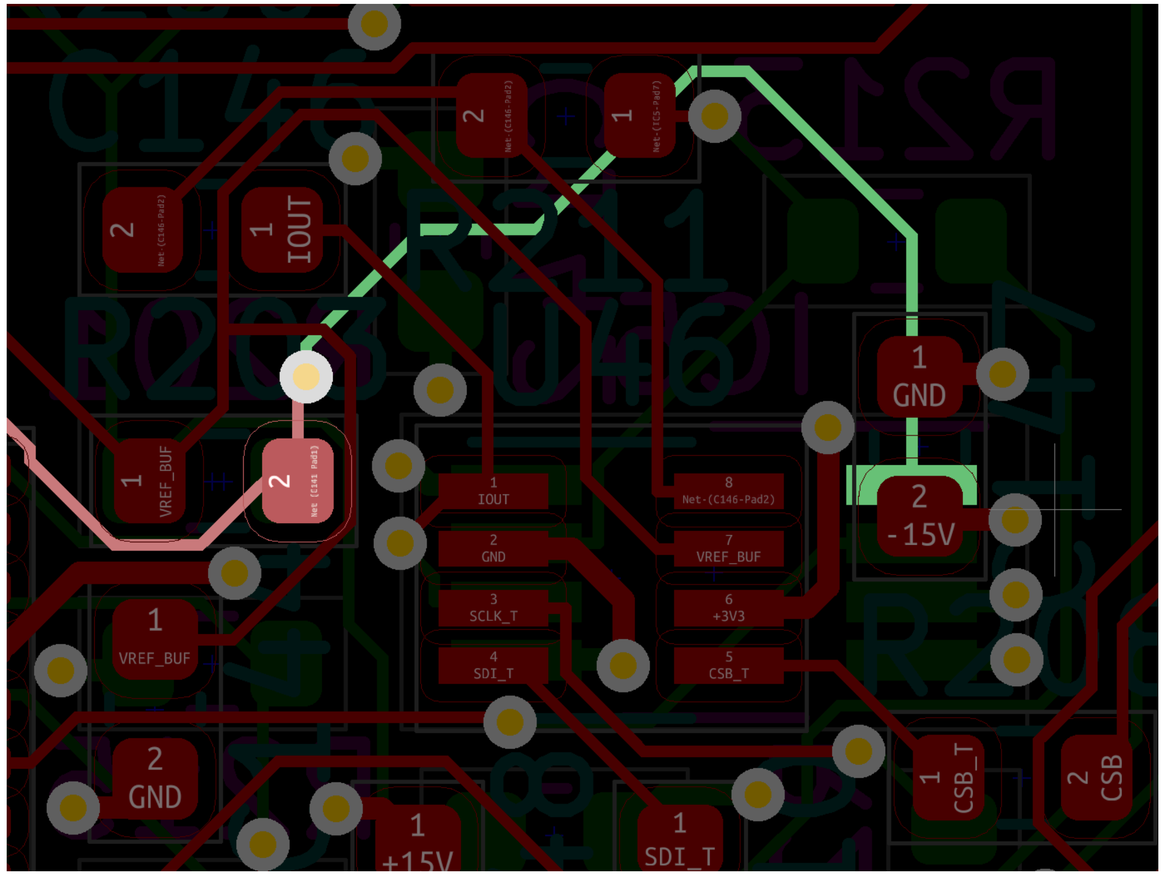


Near IOUT of DAC4
D1_SDI is in near; otherwise no aggressors.

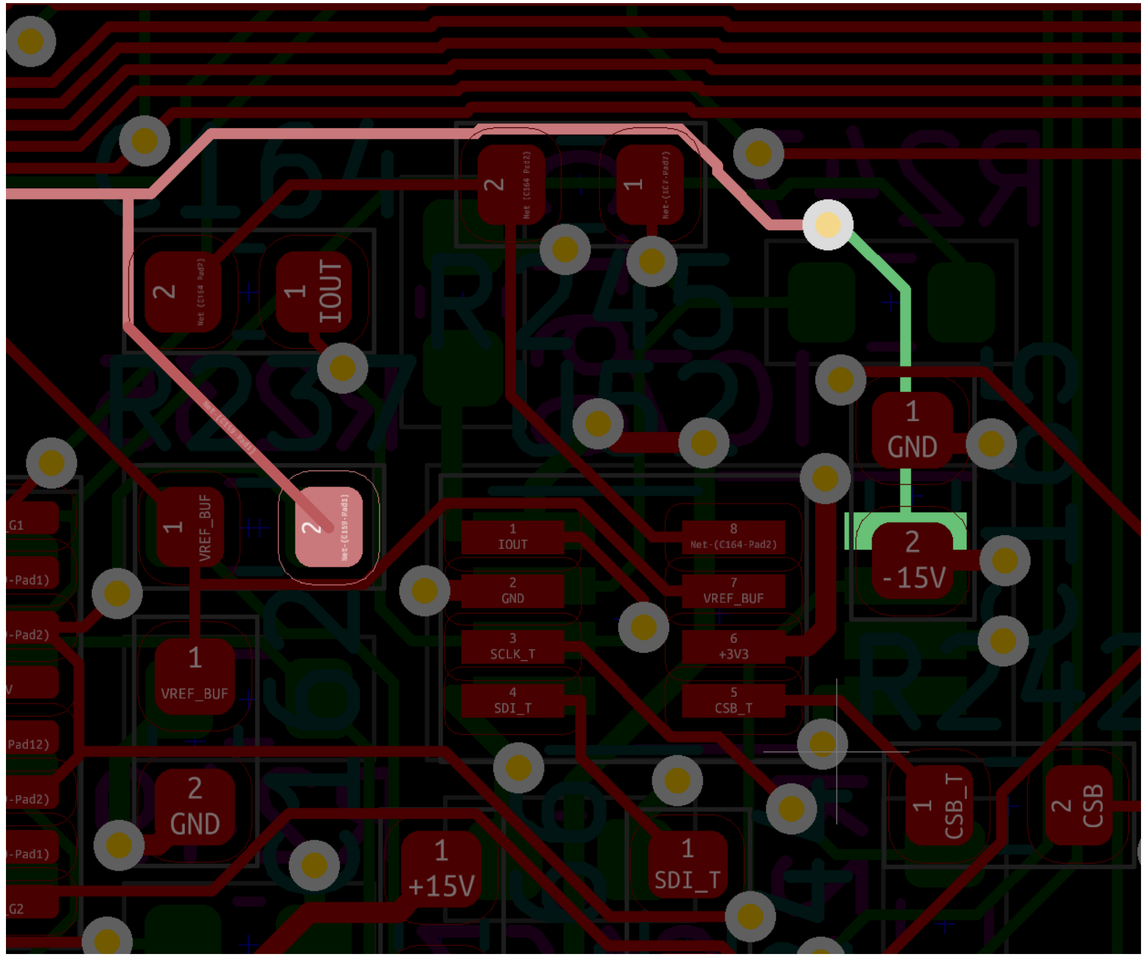


VREF_BUF after resistor is routed differently for DAC3. I think the implication is that the SPI lines are closer to IOUT on DAC3. On DAC5 the VREF_BUF after resistor effectively shields the sensitive analog signals from the SPI lines.

DAC3



DAC5



Next Steps

Measurements:

1) Determine most signifcant aggressor for Channel 5 (suspect Channel 1, D1_SDI)
2) Determine if Channel 3 and Channel 5 ripple depends on data. Set DAC to 0x0000 so that SDI is held low.

Layout/Schematic:

1) Name the high impedance nodes so that its easier to track clearances during layout.
2) Increase separation between SPI and IOUT / similar nodes.
3) Route SPI and analog DAC signals on opposite layers of the board (top vs. bottom).